SECTION-I(ii) : One or More than one

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

1. Ans. (B,C,D)

Sol. Aspartic acid, Asparagine and Glutamine are example of non-essential amino acid

2. Ans. (B,C)

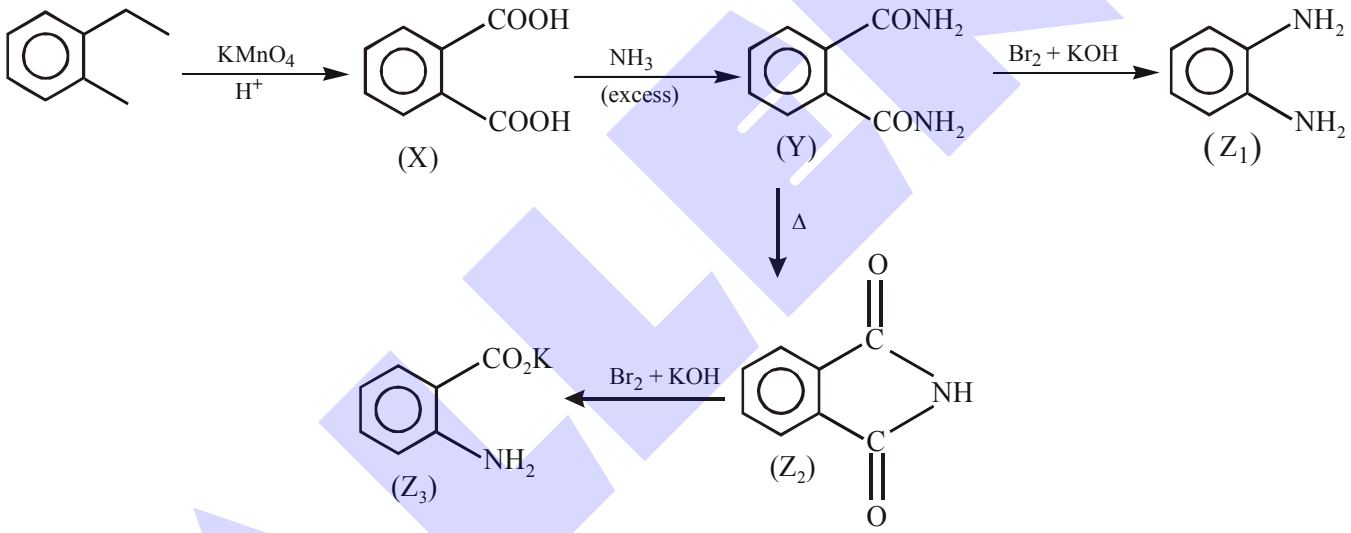
Sol. pI for alanine is 6.0

If pH of the solution is < pI it will move towards cathode

If pH of the solution is > pI it will move towards anode

3. Ans.(A,B,C,D)

Sol.



4. Which of the following polyers have vinylic monomer units?

(A) Acrilan (B) Polystyrene (C) Nylon (D) Teflon

4. Ans. (A,B,D)

Sol.

Monomer	**Polymer**
CH_2=CH–Cl Vinyl chloride	Acrlian
C_6H_5–CH=CH_2 Styrene	Polystryrene
CF_2=CF_2 Tetrafluroethylene	Teflon

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SECTION–I(iii) : Paragraph Type

This section contains **3 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

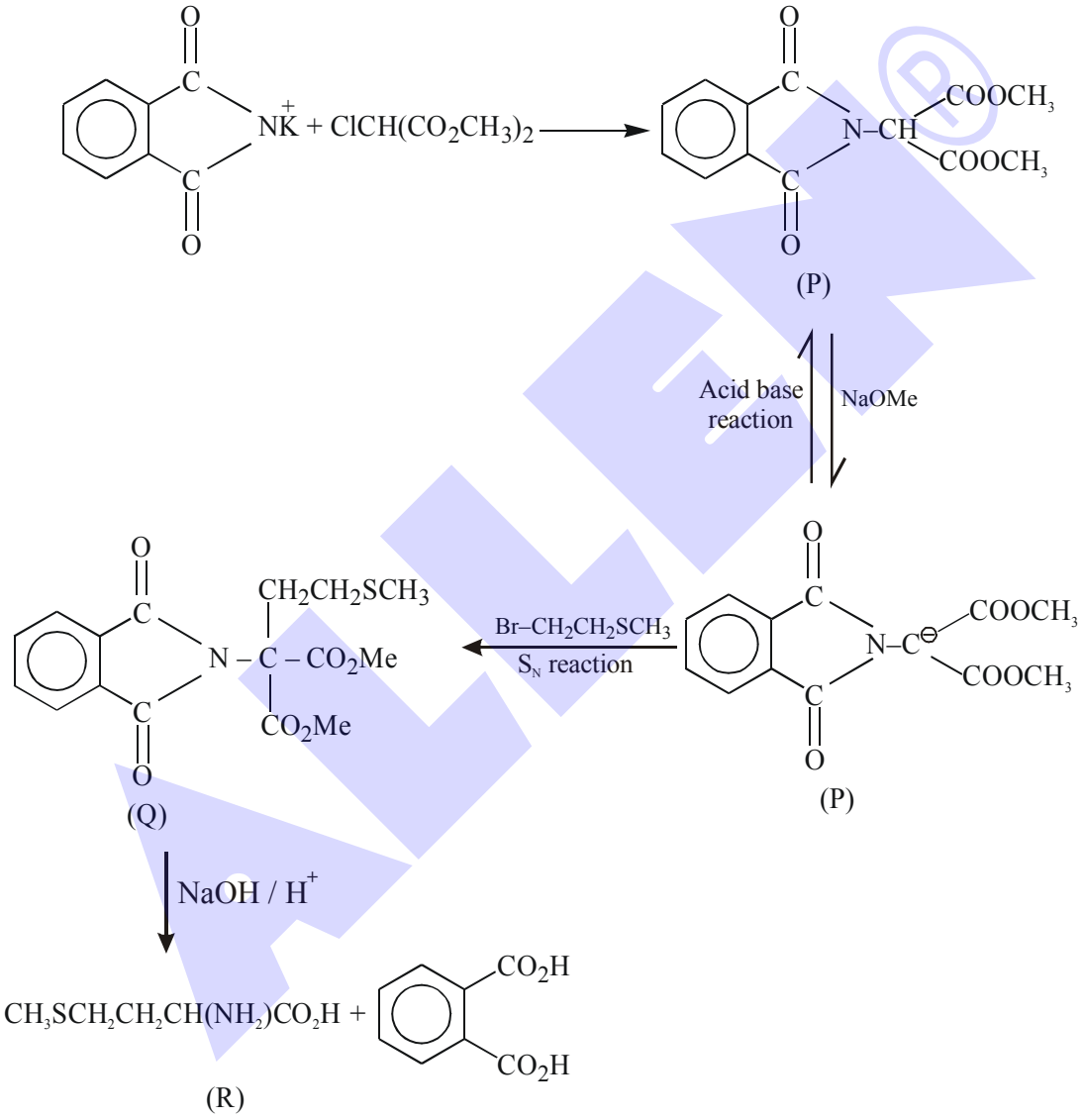
Combine solution of Q.no. 5 to 7

5. Ans.(C)

6. Ans.(B)

7. Ans.(D)

Sol.



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Combine solution of Q.no. 8 to 10

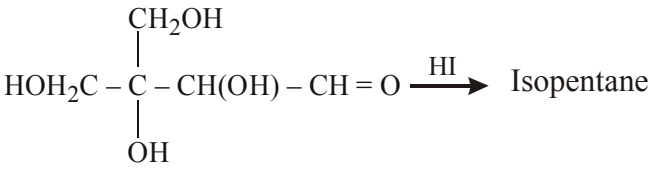
8. Ans.(C)

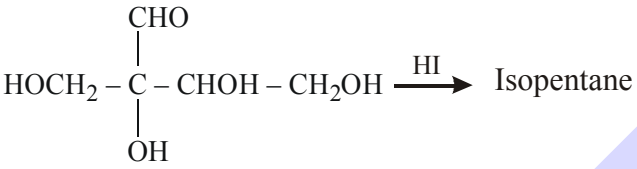
9. Ans.(C)

10. Ans.(D)

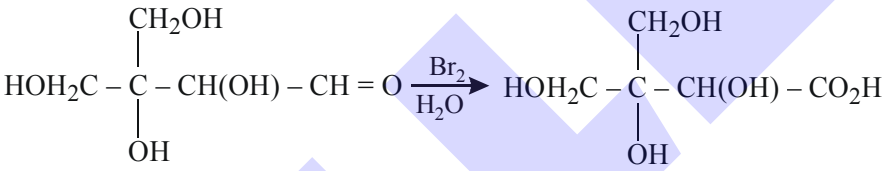
Sol. $C_5H_{10}O_5$ = aldopentose (aldehyde group should be present)

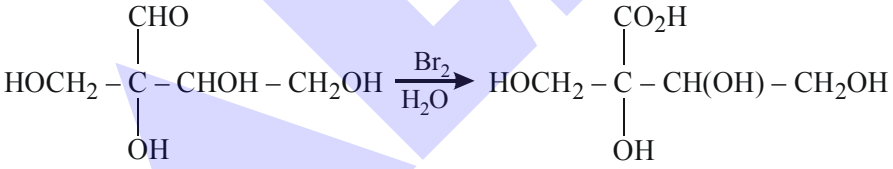
4 –OH groups must be present as it gives tetra acetate with Ac_2O





thus both structurs of A can be possible





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Combine solution for Q.no. 11 to 13

11. Ans.(C)

12. Ans.(D)

13. Ans.(B)

	(Monomer)	**(Polymer)**
(I)	$PhCH = CH_2$ Styrene	Polystyrene
(II)	$CH_2 = CHCl$ Vinyl chloride	Poly Vinyl chloride
(III)	$CH_2=C(CH_3)–CH=CH_2$ Isoprene	Natural rubber
(IV)	Caprolactum	Nylon-6

SECTION-II : (Maximum Marks: 20)

1. Ans. (0.00)

Sol. $Na^+[CO_2^- - CH_2CH_2CH(NH_3^+)CO_2^-]$

2. Ans. (10.00)

Sol.

3. Ans.(4.00)

Sol.

4. Ans. (1.00)

Sol. polypentoside (pentosan) is polymer of pentose.

$(C_5H_{10}O_5 - H_2O)_n \Rightarrow (C_5H_8O_4)_n$

$a + c - b = 5 + 4 - 8 = 1$

5. How many of the following compounds will give silver mirror with Tollen's reagent

MeCHO, PhCHO, PhNHOH, Propyne, $MeCO_2H$, HCO_2H, Sucrose, Starch, α -D-Glucopyranoside

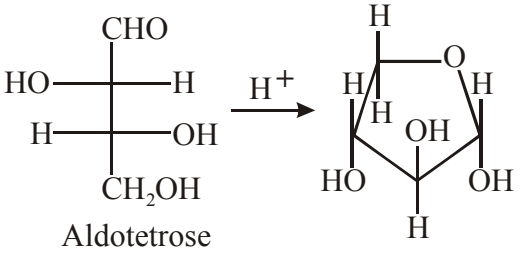
5. Ans. (4.00)

Sol. MeCHO, PhCHO, PhNHOH, HCO_2H

6. The number of carbon atoms in smallest aldose able to for a cyclic stable hemiacetal

6. Ans. (4.00)

Sol.



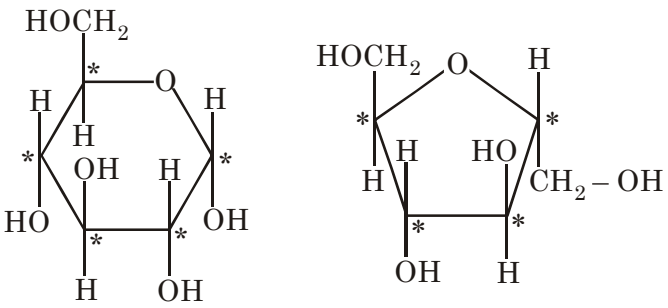
Aldotetrose

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7. If x is the number of chiral centre in α–D–glucopyranose and y is the number of chiral center of β-D-fructofuranose then find out sum of x + y = ?

7. Ans. (9.00)

Sol.



8. Ans.(5.00)

Sol. It is example of step-up reaction

SECTION–IV : Matrix Match

This section contains **FIVE** questions.

1. **Ans. A–P,R, S; B–P,S; C–P,R; D–Q,S,T**

Sol. (A) Ala gly phen ala $\xrightarrow{\text{Hydrolysis}}$ Glycine + Alanine + Phenylalanine
Tripeptide linkage

(B) Val Leu isoleu $\xrightarrow{\text{Hydrolysis}}$ Valine + Leucine + Isoleucine
Tripeptide linkage

(C) Gln Gly Glu $\xrightarrow{\text{Hydrolysis}}$ Glutamine + Glycine + Glutamic acid
Tripeptide linkage

(D) Ala Asn Asp Gly $\xrightarrow{\text{Hydrolysis}}$ Alanine + Asparagine + Aspartic acid + Glycine
Tetrapeptide linkage

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2. Ans. A–R; B–P,R; C–Q,S,T; D–P,R

Sol. Carbohydrate ⇒ Polyhydroxy aldehyde or ketone

Ninhydrin test ⇒ Test for α-amino acid

3. Ans. A–R,T; B–S,T; C–Q,T; D–P, T

Sol. Lexan ⇒ polycarbonate

Kevlar ⇒ Polyamide

Plexiglass ⇒ Polymer of methyl methacrylate

Neoprene ⇒ Polymer of 2-chloro 1,3-butadiene

All above are examples of Synthetic polymer

4. Ans. A–Q,T; B–R; C–P,T; D–S, T

Sol. Novolac ⇒ Polymer of Phenol and formaldehyde resin

Glyptal ⇒ Polyester of glycol and phthalic acid

Buna–S ⇒ Copolymer of 1,3–butadiene and styrene

Dacron ⇒ Polyester of glycol and terephthalic acid

5. Ans. A–P, R; B–Q; C–Q, S; D–Q,R,S,T;

Sol. (A) Guta–purcha ⇒ Chain growth polymer, Natural polymer, Biodegradable polymer

(B) Glyptal ⇒ Step growth polymer

(C) Nylon-2-nylon-6 ⇒ Step growth polymer, Biodegradable polymer

(D) Cellulose ⇒ Step growth polymer, Natural polymer, Biodegradable polymer, Glycosidic linkage

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